

Task Questions

1. Question 1

Answer:

An internet gateway is a horizontally scaled, redundant, and highly available VPC component. It allows both inbound and outbound traffic to be passed from and out of the AWS network to the internet. It also performs network address translation (NAT) for instances that were assigned public IPv4 addresses

Where as on the other hand, NAT gateway only allows the private subnet to access the internet while hiding its private IP address. It only supports outbound traffic to be passed from private subnet to the internet and prevents the inbound traffic to the private subnet from the internet.

2. Question 2

Answer:

First, we have to create the Internet Gateway (IGW), this allows communication between the instances in our AWS network and the internet. here we create it and give it a name like, Lab IGW.

After that, we will have to attach the internet gateway to the VPC, which will establish a connection between the VPC and the internet.

now it is time to create a route table which contains rules for directing network traffic within the VPC. we should associate this route table with our VPC.

Then, we will have to edit the routes in the public route table to include a route that sends all the traffic to the internet gateway we created earlier.

At last associate the newly created public route table with the subnet that we want to make public. This way any instance launched in this subnet will use the route table's rules for the internet traffic.

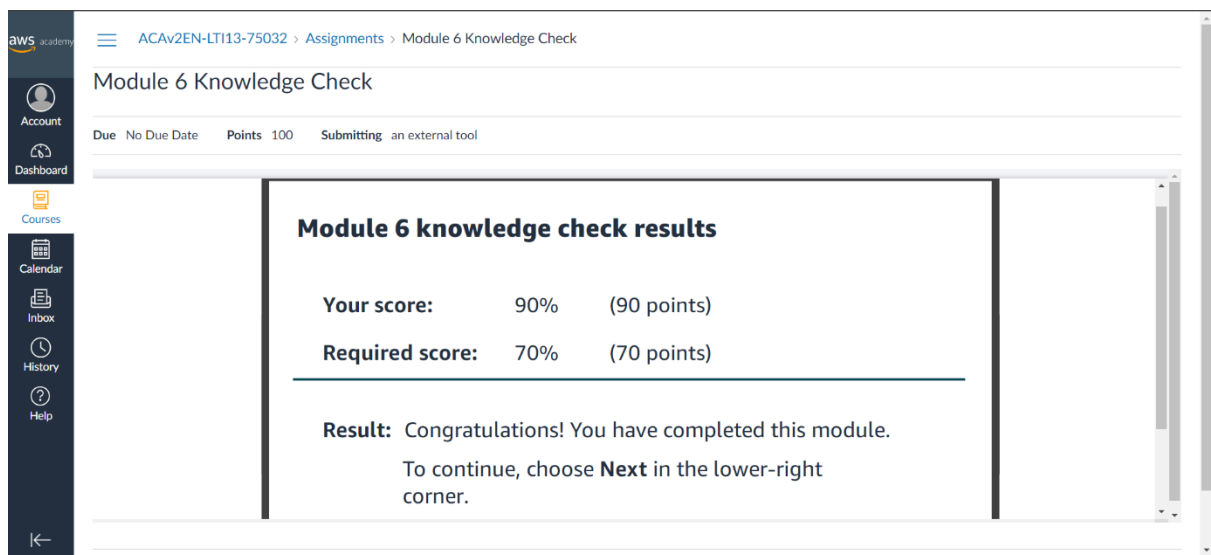
This is how we make a subnet public.

Quiz

1. Knowledge Check for Module 6

This knowledge check was comprised of the most important questions throughout the module 6. The questions asked was kind of a summary to the whole module 6. In this knowledge check I was able to see my understanding of this module, and the from the mistakes I was able to correct myself of the parts that I was not sure about.

This knowledge check was mainly about the Virtual Private Cloud. The first question itself was asking about its definition. Then there were questions about the best practices to use when implementing the VPC. After that, most of them were about scenario-based questions, what will I do to solve the problem in that situation. Then how to implement the security groups and access control lists to securely use it. Overall I was able to learn a lot from this quiz itself.



The screenshot shows the AWS Academy interface for a knowledge check. On the left is a dark sidebar with navigation icons for Account, Dashboard, Courses, Calendar, Inbox, History, and Help. The main content area has a breadcrumb trail: ACAV2EN-LT113-75032 > Assignments > Module 6 Knowledge Check. Below this is the title 'Module 6 Knowledge Check' and a status bar showing 'Due: No Due Date', 'Points: 100', and 'Submitting: an external tool'. The central part of the page displays the results in a white box with a black border. It shows a score of 90% (90 points) out of a required 70% (70 points). A green progress bar is visible. The result message says 'Congratulations! You have completed this module. To continue, choose Next in the lower-right corner.'

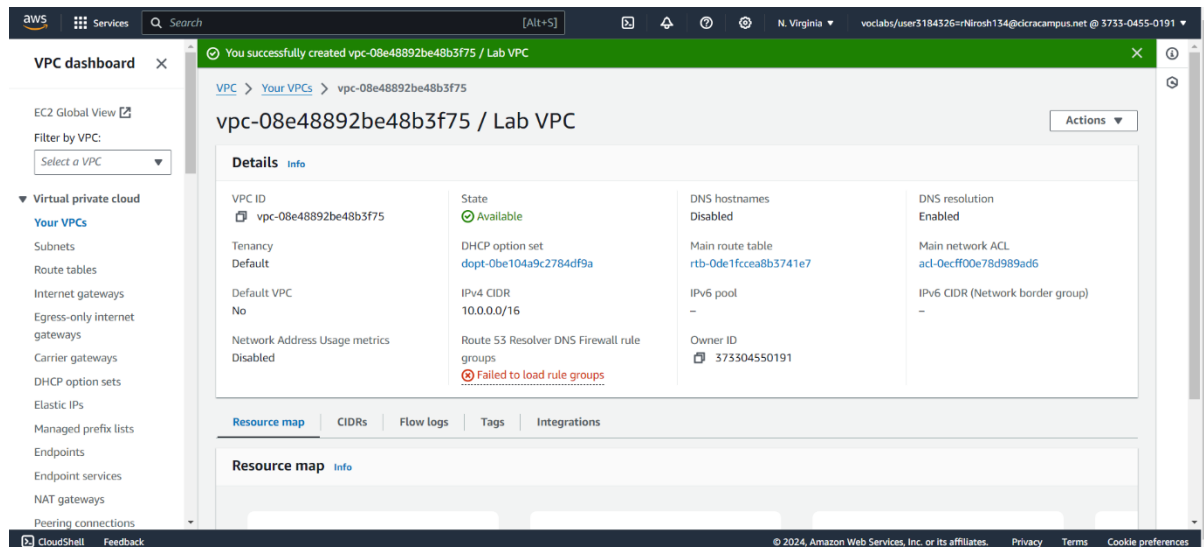
Module 6 knowledge check results		
Your score:	90%	(90 points)
Required score:	70%	(70 points)
Result: Congratulations! You have completed this module. To continue, choose Next in the lower-right corner.		

Module 6 Guided Lab - Creating a Virtual Private Cloud

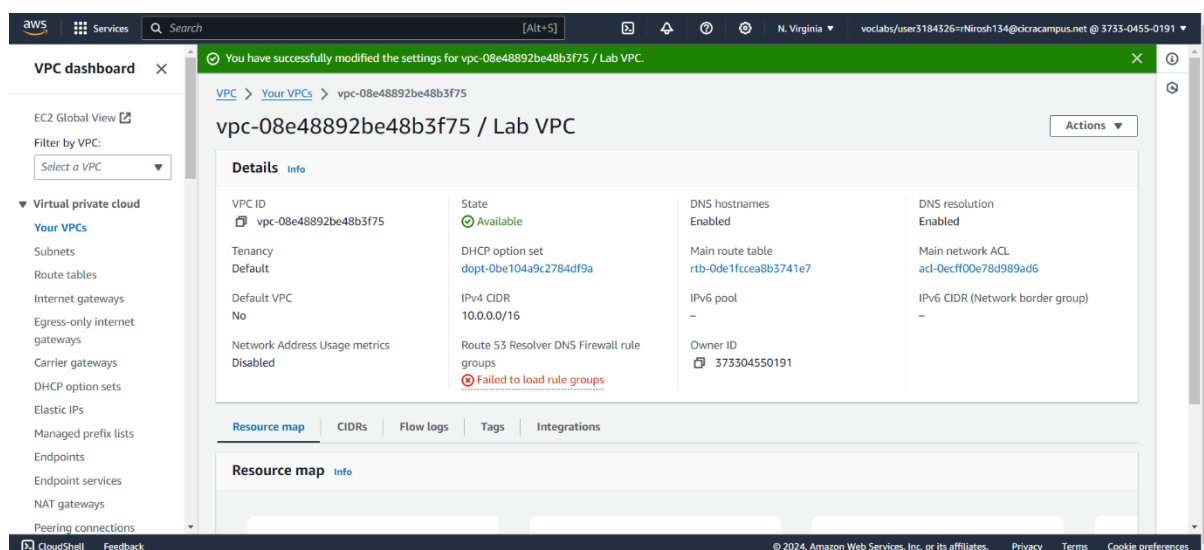
Task 1: Creating a VPC

First, I accessed the VPC console in AWS to create and manage the VPC.

I give it a name "Lab VPC", an ip address range or CIDR block is assigned (10.0.0.0/16) which gives 65000 IP addresses to work with.



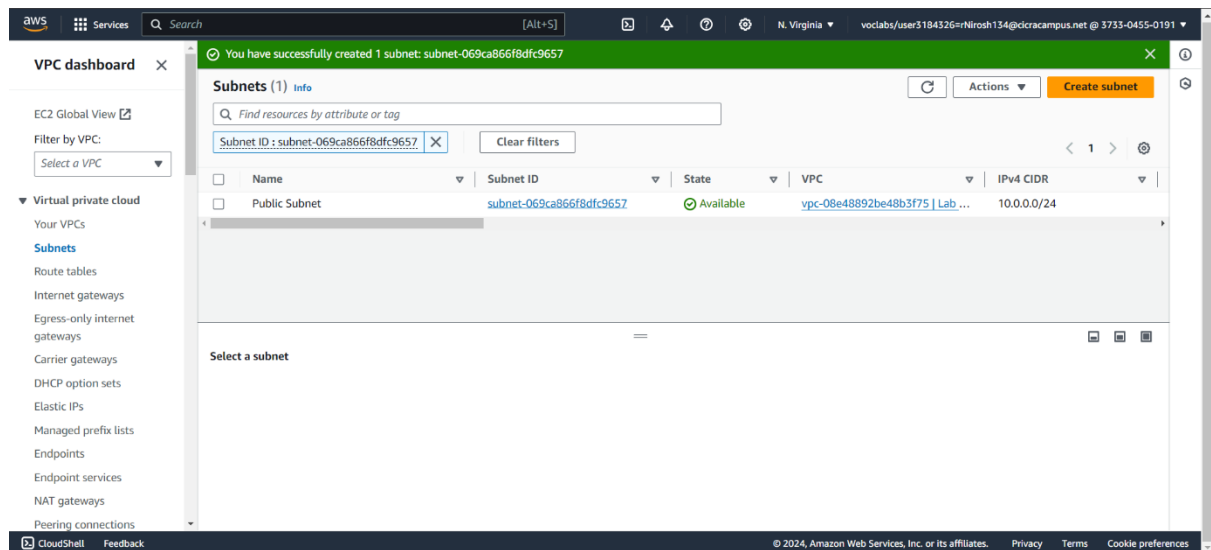
Then I enabled the DNS hostnames, any EC2 instances launched within the VPC will automatically get a DNS hostname. This will make it easier to identify and connect instances over the internet.



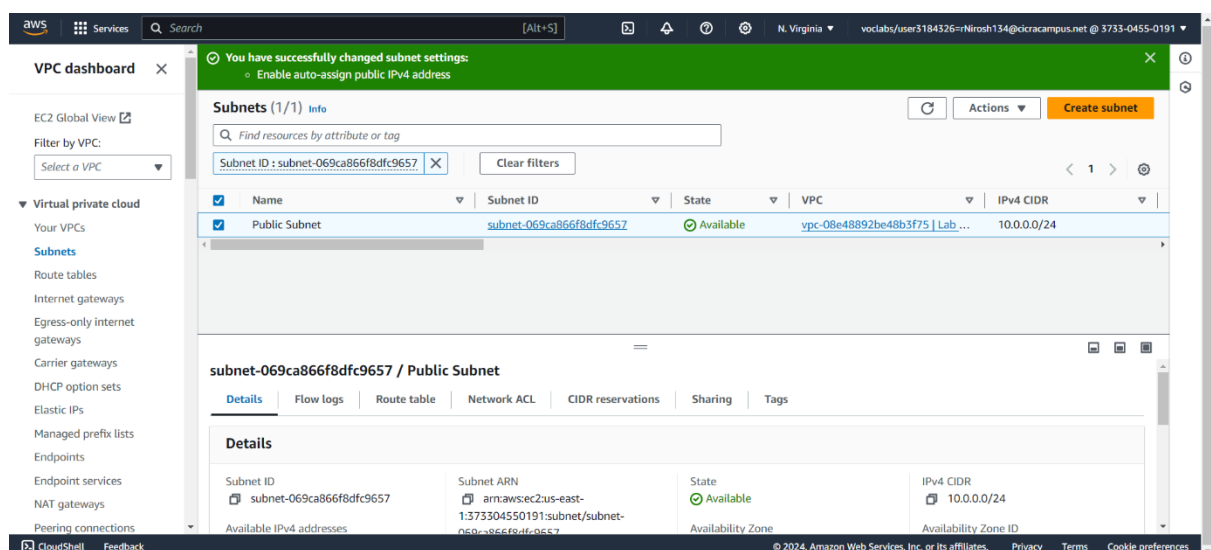
Task 2: Creating subnets

In this task we create both a public and private subnet.

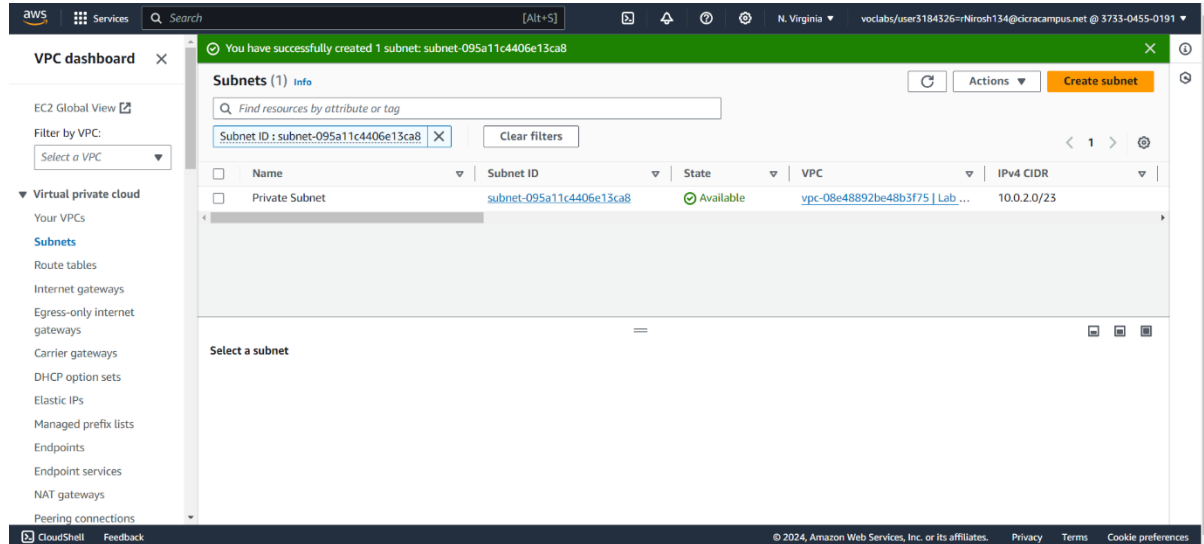
First, I create the public subnet for resources that need to connect to the internet. Here we are configuring the VPC ID, choose the availability zone for the subnet with high availability. At last defined the range of IP addresses available for resources within it.



After creating the public subnet, I'm configuring it to automatically assign a public IP address to any instance launched within it.



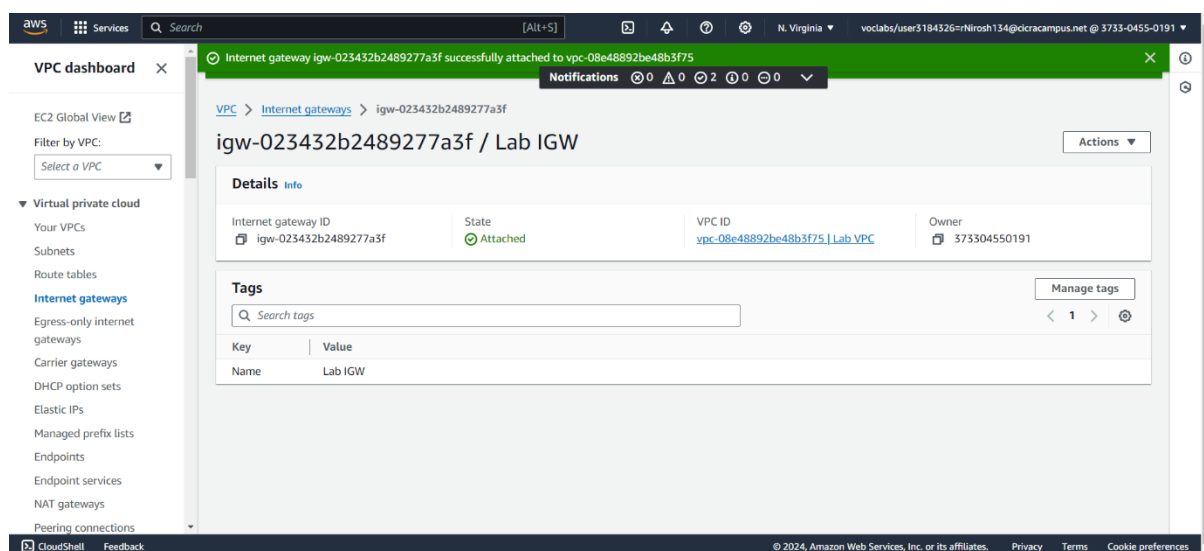
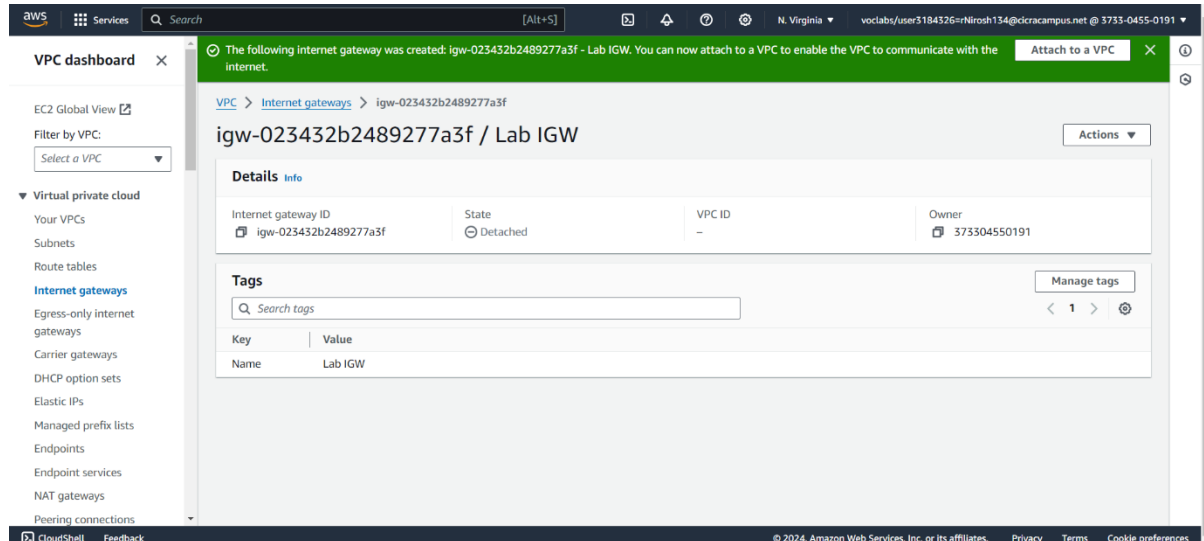
Then created the Private subnet for the resources that need to remain isolated from the internet. similar configurations are done to this as well, however the CIDR block is larger (10.0.2.0/23) to accommodate more resources.



Task 3: Creating an internet gateway

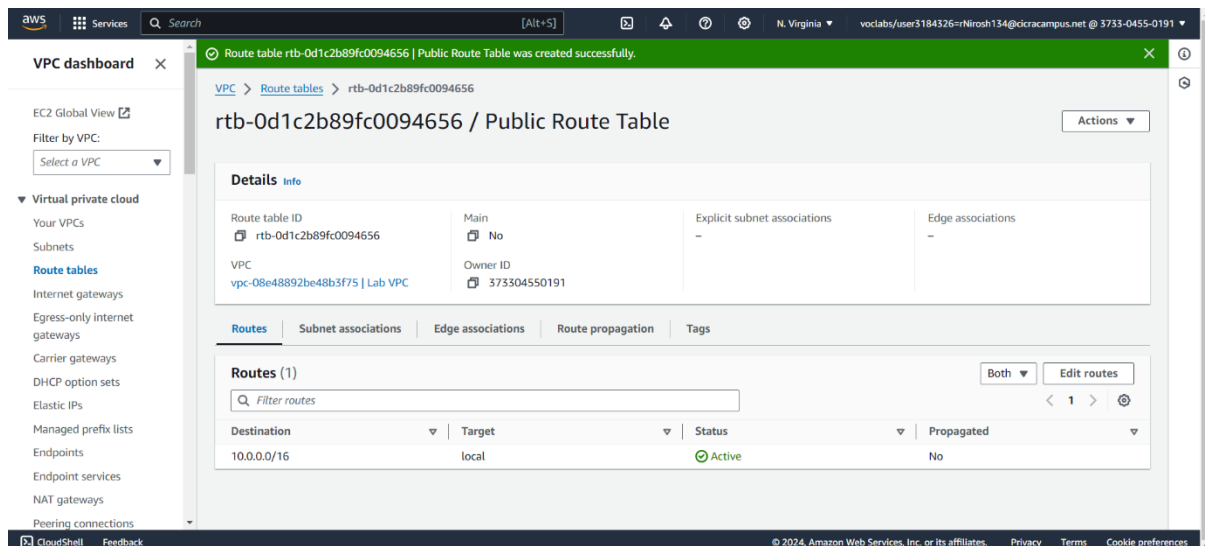
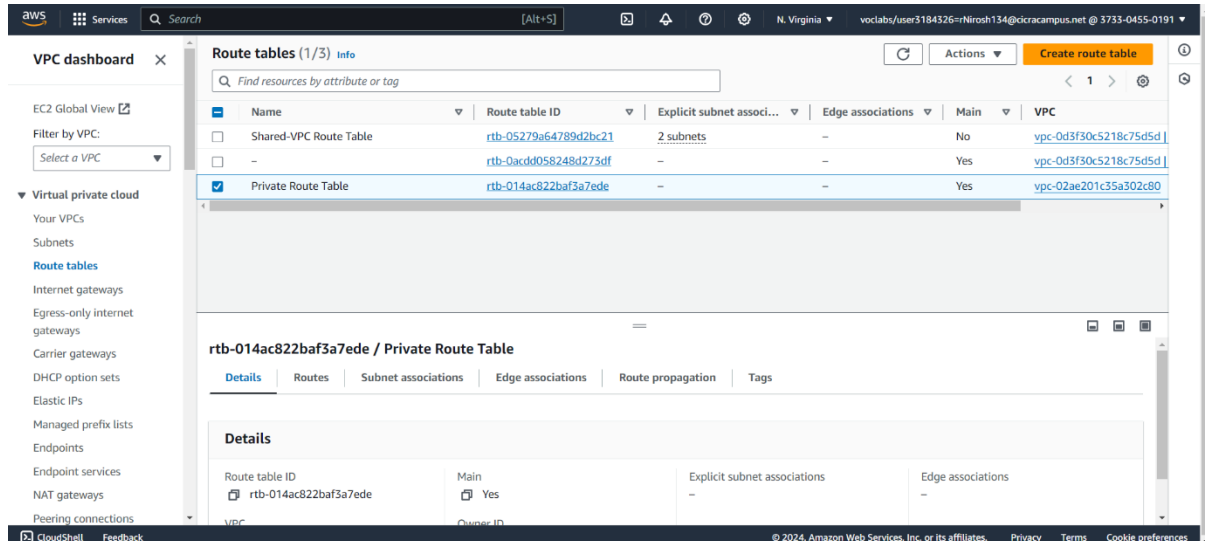
Now I'm going to the internet gateways option in the VPC dashboard. A new Internet gateway is created and given the name "Lab IGW"

Then the internet gateway is attached to the VPC to enable internet connectivity for resources within the VPC. Configurations like desired VPC are selected and attached.

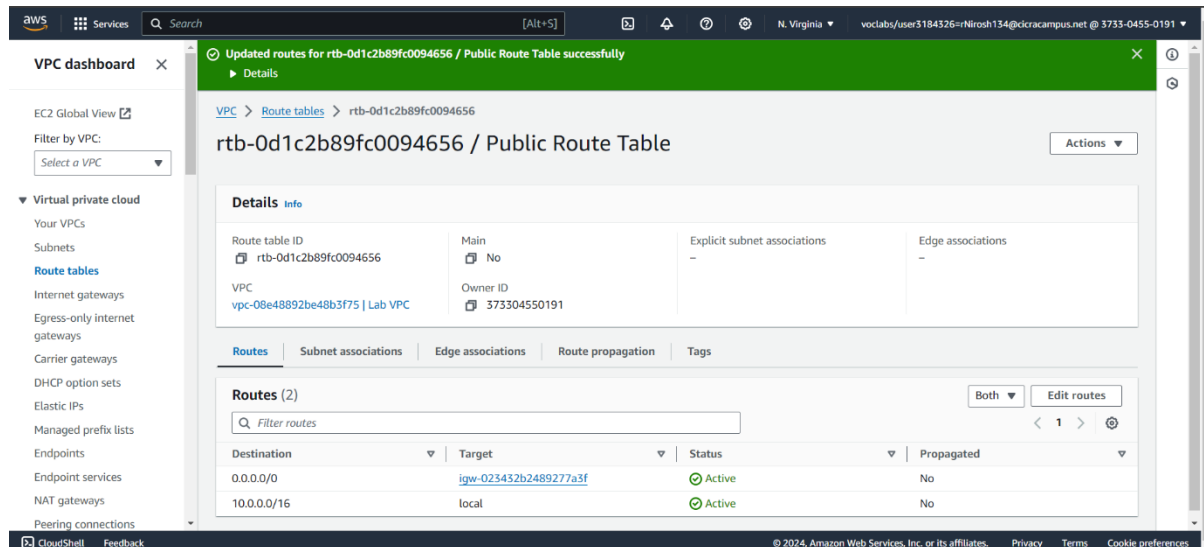


Task 4: Configuring route tables

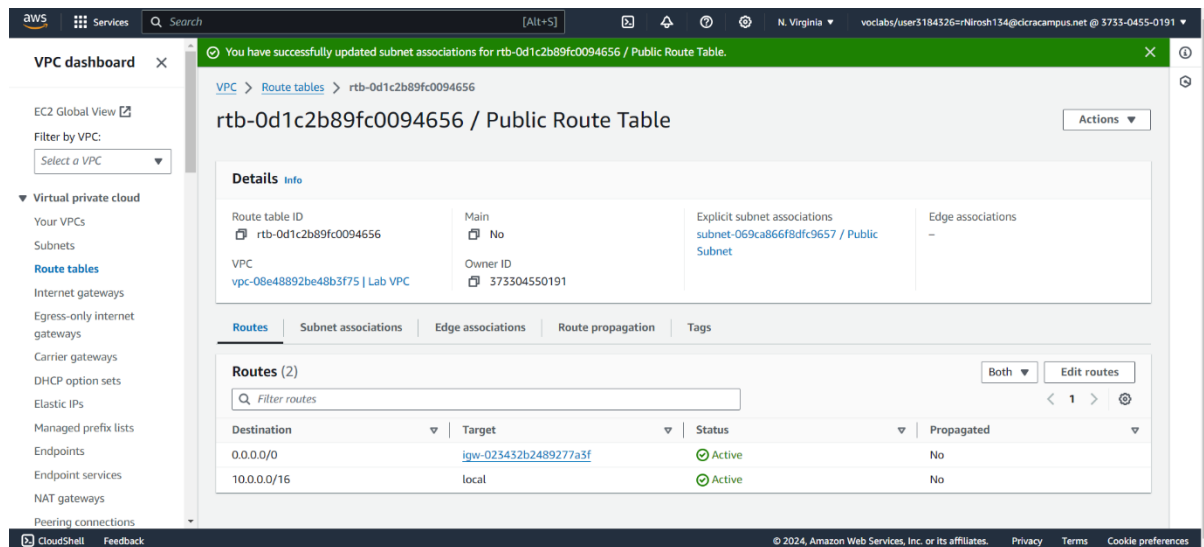
A new route table called "public route table" is created for directing internet bound traffic. specifying the VPC that this route table belongs to.



Then configuring the routes, like adding rule that directs internet-bound traffic to the internet gateway that was created earlier.



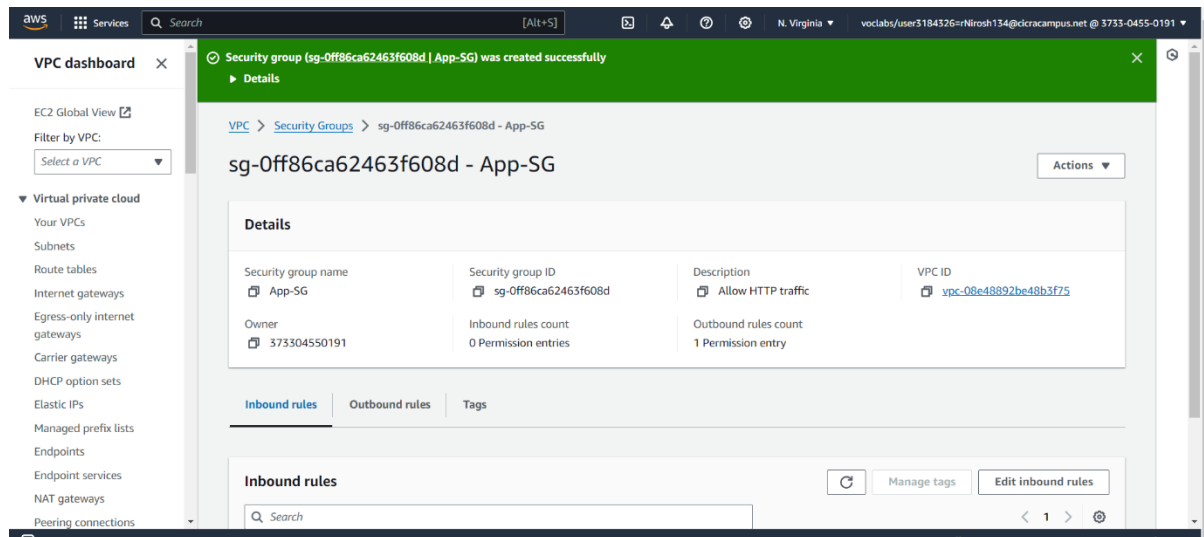
Finally the newly created public route table is associated with the public subnet created before.



Task 5: Creating a security group for the application server

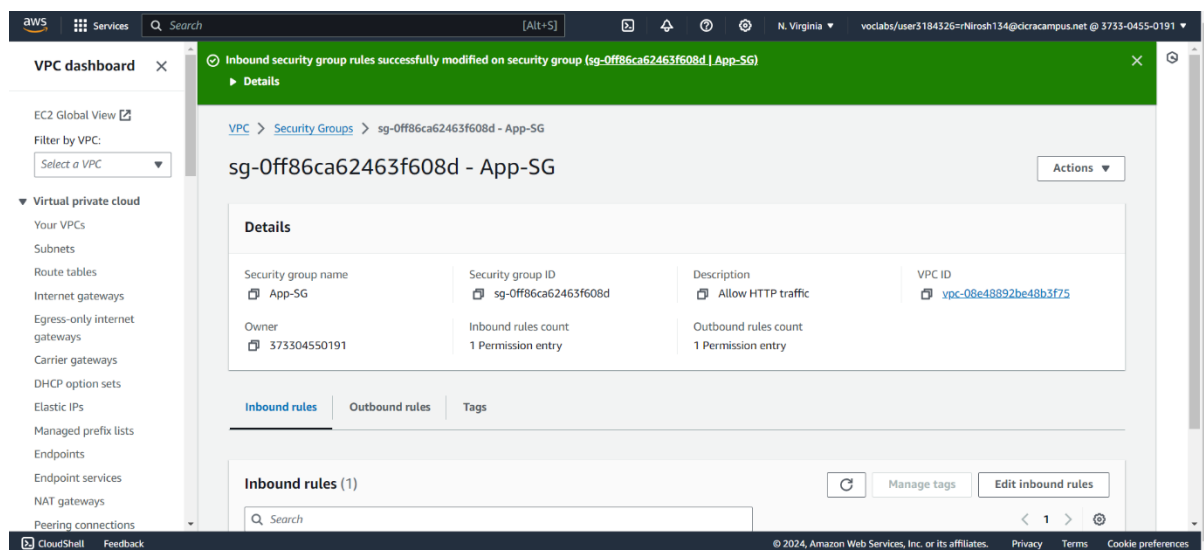
Here we are creating a new security group called "App-SG" to control access to the application server.

Then we give a description and select a VPC to associate this security group with.



Then we configure some inbound rules.

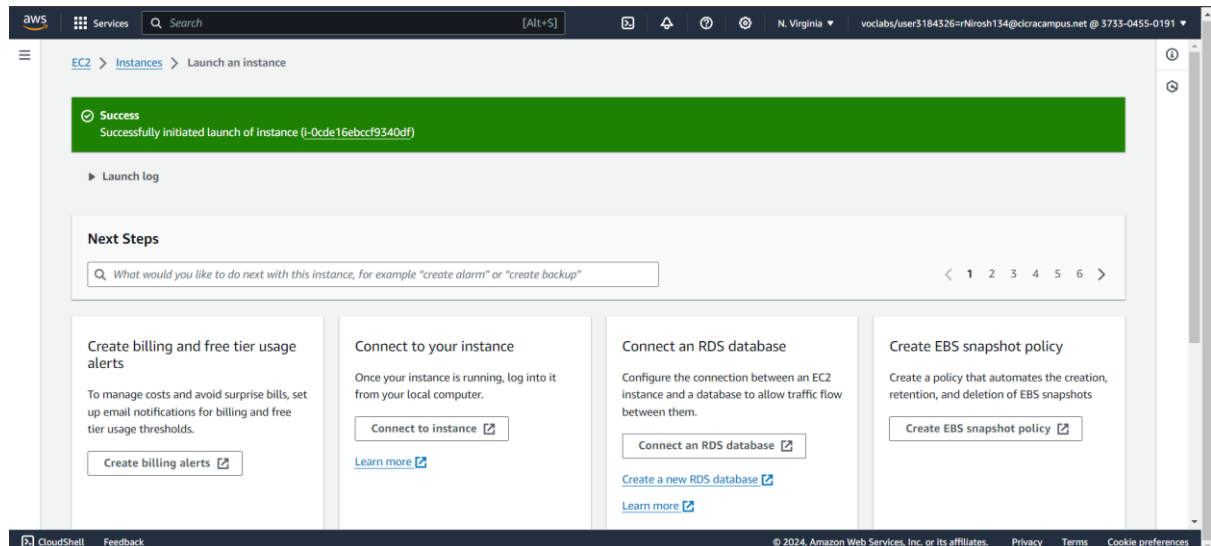
Finally, we add an inbound rule for the HTTP traffic



Task 6: Launching an application server in the public subnet

Now at last we are launching the new EC2 instance and configure its setting.

after creating it we need to wait for a few minutes until the status shows 2/2 checks passed indicating it's initialized and ready.



Once the instance is ready, we copy the Public IPv4 DNS value, then paste that address in a new browser tab to access the EC2 instance.

Then the page loads up and confirms that the public subnet was configured properly.



Grade:

My grade is 45/45

The screenshot displays the AWS Academy interface for a completed lab. On the left is a dark sidebar with navigation icons for Account, Dashboard, Courses, Calendar, Inbox, History, and Help. The main content area is titled "Module 6 Guided Lab - Creating a Virtual Private Cloud". Below the title, it shows "Due: No Due Date", "Points: 100", and "Submitting: an external tool". A toolbar includes buttons for "Submit", "Details", "AWS", "Start Lab", "End Lab", "Instructions", "Grades", and "Actions". The "Grades" tab is active, showing a "Total score" of 45/45. A list of tasks is displayed on the right: [Task 1] Lab VPC CIDR, [Task 2A] Public Subnet, [Task 2B] Auto-Assign Public IP, [Task 2C] Private Subnet CIDR, and [Task 3] Lab VPC has IGW. A terminal window in the center shows a successful login to an AWS instance. The submission details on the far right indicate the lab was completed on May 4 at 7:31pm with a grade of 100/100.

aws academy

ACAv2EN... > Assignments
> Module 6 Guided Lab - Creating a Virtual Private Cloud

Module 6 Guided Lab - Creating a Virtual Private Cloud

Due No Due Date Points 100 Submitting an external tool

Submit Details AWS Start Lab End Lab Instructions Grades Actions

Files README Terminal Source

EN_US

Module 6 - Guided Lab: Creating a Virtual Private Cloud

Lab overview and objectives

Traditional networking is difficult. It involves equipment, cabling, complex configurations, and specialist skills. Amazon Virtual Private Cloud (Amazon VPC) hides the

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Total score 45/45

- [Task 1] Lab VPC CIDR
- [Task 2A] Public Subnet
- [Task 2B] Auto-Assign Public IP
- [Task 2C] Private Subnet CIDR
- [Task 3] Lab VPC has IGW

Submission

May 4 at 7:31pm
[Submission Details](#)

Grade: 100 (100 pts possible)
Graded Anonymously: no

Comments:
No Comments